◆ TRACK 2 — MACHINE LEARNING

The Databricks ML Platform, End to End — with Genie Code

Attendees build a claim-case escalation model across every layer of the Databricks Data Science & ML platform, using Genie Code for ML as their AI pair-programmer — from EDA to a monitored, governed serving endpoint deployed as code.

DURATION	ATTENDEES	FACILITATOR	REVIEWER
3.5 hrs · 2 breaks	~12	SA + Inspire11	Jack Law

REPO
saswata30/Allianz_ML_Workshop

01 What attendees walk away able to do

Build with Genie Code

Plan, write and run ML code from the Genie Code panel — EDA, features, XGBoost training, predictions — while reviewing every step.

Engineer governed features

Publish a Unity Catalog feature table and build training sets with lineage from feature to model.

Track, register & serve

Compare runs in MLflow, register the champion in Unity Catalog, score in batch and deploy a serverless endpoint.

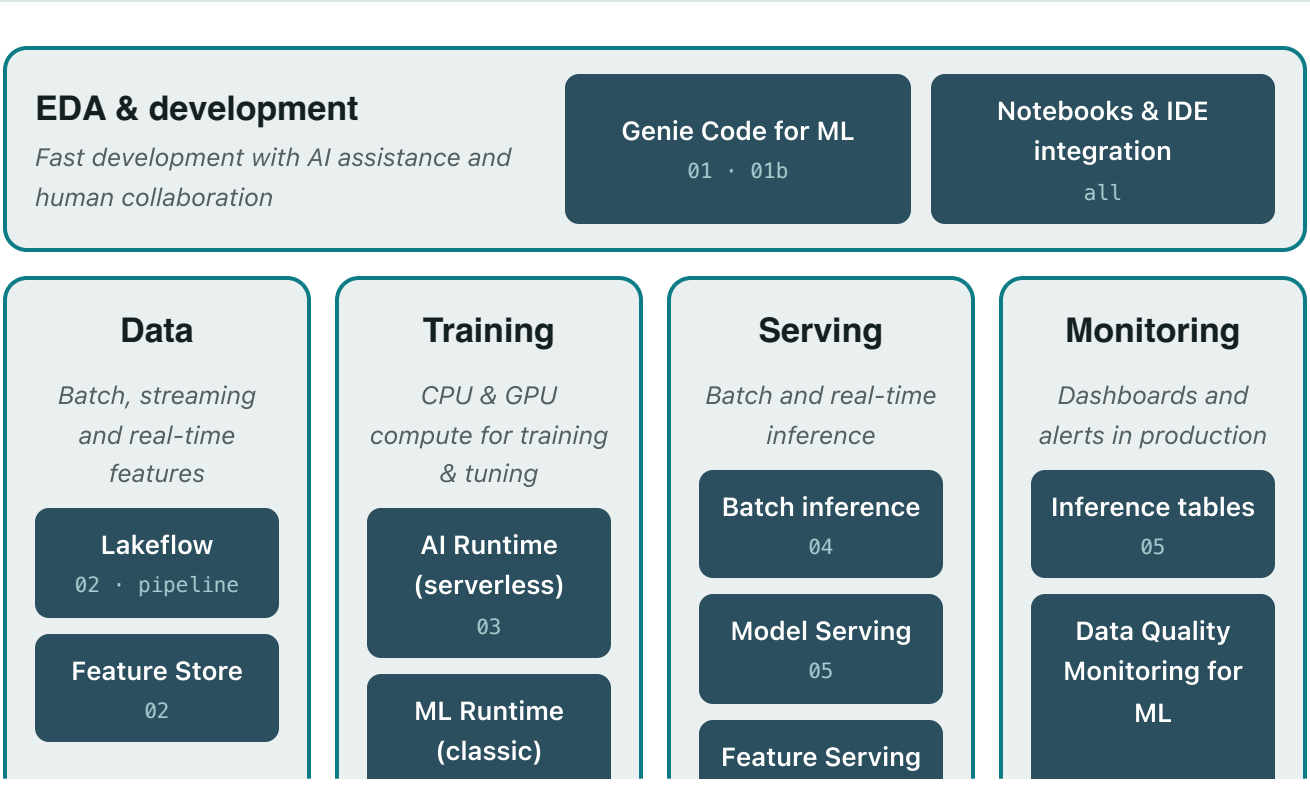
Monitor & operationalise

Watch quality and drift with Data Quality Monitoring, and run the whole pipeline as scheduled jobs with champion/challenger promotion.

---

02 The platform map — and where each module lands

The workshop walks the standard Databricks Data Science & ML platform. Module numbers refer to notebooks in the repo.



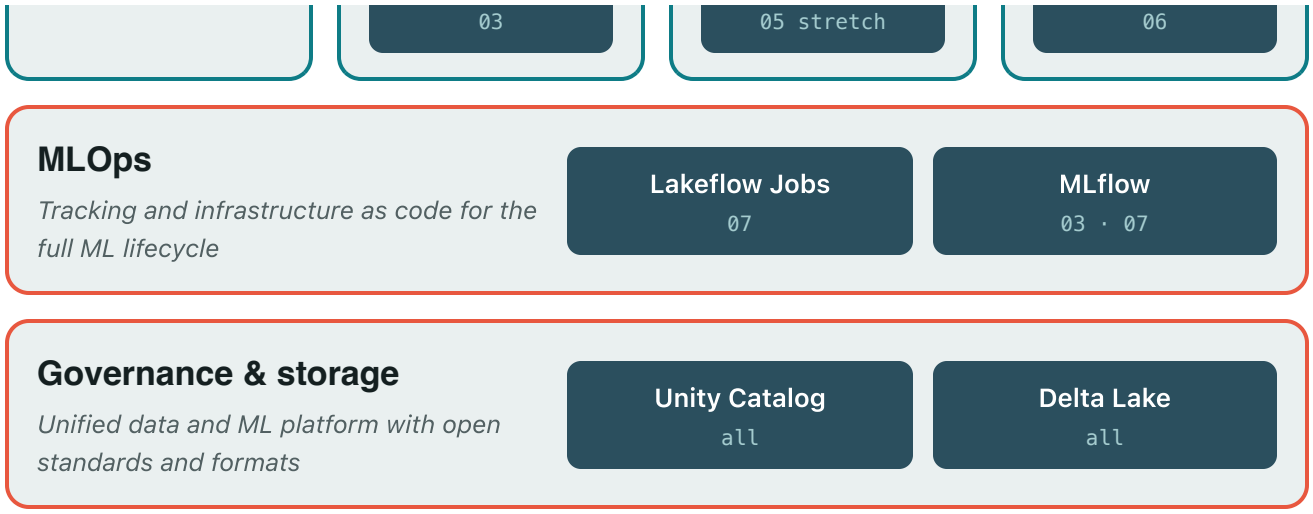


Figure 1 — Every layer, one use case. Genie Code accelerates the work at each layer; Unity Catalog governs every table, feature, model and endpoint; MLflow records what produced each model. Genie Code for ML replaces AutoML for customised model development.

03 Assumptions & prerequisites

WORKSPACE (CONFIRM WITH ALLIANZ)

- [x] **Partner-powered AI features** enabled for the account and workspace (required for Genie Code).
- [x] Catalog `adp_workshop` with `USE CATALOG` + `CREATE SCHEMA` for attendees.
- [x] Serverless notebooks and jobs (or DBR 16.4 LTS ML+ shared cluster).
- [x] Model Serving and Data Quality Monitoring available in the region.
- [x] Optional: curated `adp.claims.cases` table —

AT EACH SEAT

- [x] Git folder created from `github.com/saswata30/Allianz_ML_Workshop`.
- [x] `00_setup_data` run once — personal schema `ml_<user>` with 60k labelled and 3k open cases.
- [x] Genie Code panel opens and **Agent** mode is available.
- [x] A small serverless SQL warehouse for the monitoring dashboard and alert.

otherwise the synthetic data is used.

04 Agenda — 3.5 hours

Timeboxes assume a 09:00 start. ▸ HANDS-ON marks build blocks. Each block gives a Genie Code prompt first and a tested reference solution second.

0:00–0:15 **Welcome, use case & platform map** 15m
The escalation question, the data, and Figure 1.

0:15–0:25 **00 · Setup** 10m ▸ HANDS-ON
Git folder, serverless, personal schema and synthetic cases.

0:25–0:50 **01 · EDA with Genie Code** 25m ▸ HANDS-ON
Profile, segment and visualise with `@cases`; ask Genie Code to propose features.

0:50–1:10 **01b · Genie Code challenge** 20m ▸ HANDS-ON
One prompt → plan → explore, features, XGBoost, predictions. Debrief: what's missing for production?

1:10–1:35 **02 · Features & Feature Store** 25m ▸ HANDS-ON
Build `case_features` with a primary key; labels kept separate; Lakeflow pipeline walkthrough.

1:35–1:50 ☕ BREAK 15M

1:50–2:30 **03 · Train, track & register** 40m ▸ HANDS-ON
FeatureLookup training set; logistic regression vs gradient boosting vs XGBoost in MLflow; register `@champion` in Unity Catalog.

2:30–2:50 **04 · Batch inference** 20m ▸ HANDS-ON
Spark-UDF scoring by alias into `case_predictions`; who to call first.

2:50–3:00 ☕ BREAK — ENDPOINTS BUILD 10M

3:00–3:15

05 · Model Serving 15m ▶ HANDS-ON

Serverless endpoint with AI Gateway inference table; query it; Genie Code endpoint health check.

3:15–3:30

06 · Monitoring · 07 · Jobs & lifecycle · recap 15m

Drift on a simulated storm surge; training and scoring jobs plus a Lakeflow pipeline created from a notebook; champion/challenger; next steps.

---

05 Actions & sign-off

→ ALLIANZ

Confirm the workspace

Partner-powered AI features, catalog grants, serverless, Model Serving and monitoring availability.

→ ALLIANZ

Choose the data

Synthetic cases (default) or the curated `adp.claims.cases` table with the same columns.

→ DATABRICKS

Dry-run the build

Run 00 → 07 once with a test user; time-box each block; pre-create one endpoint as a fallback.